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Proceedings of the Combustion Institute 33 (2011) 649–657
Proceedings of the Combustion Institute
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Surface enhanced Raman spectroscopy (SERS) of particles produced in premixed flame across soot threshold

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Available online 6 October 2010

Abstract

Surface enhanced Raman scattering (SERS) has been employed for the first time to characterize particles produced in flames across the soot inception.

The strong enhancement of Raman signal produced by SERS enabled the measurement of the Raman spectra of size-selected particles collected by a dilution probe-DMA system and deposited by electrophoresis in very low concentrations on SERS substrates. It also enables investigation of second order 2*D* Raman bands in soot and soot precursors, providing complementary information to the first order Raman bands.

Raman spectra, together with the measurement of particle size distribution by DMA, help to characterize particles produced across the soot threshold limit. Three particle modes have been identified: flames with C/O < 0.65 produce only mode_I particles with a diameter of $d \approx 2$ nm, which present the typical bands of amorphous carbon and have a very low, if any, three-dimensional ordering of graphitic planes. Other aromatic macromolecules smaller than the lower detection limit of the DMA ($d < 1.5$ nm) are present in the flame, and are collected by thermophoretic sampling. In flames with higher C/O ratio the size distribution becomes bimodal with the formation of mode_II particles, $d \approx 4$ nm. In non-sooting flames, the SERS spectra of mode_I and mode_II particles are similar except for features indicating the presence of a higher degree of disorder in mode_II, likely due to distorted or five order rings.

In soot-forming flames, larger mode_III particles collected with a diameter of 40 nm present a higher ordering and a reduction of graphitic interplane distance, which, nevertheless, still remains larger than 3.50 Å. Two types of particles were observed in smaller mode_II particles collected from sooting flame conditions: one with Raman spectra similar to the mode_III particles and the other with spectra similar to particles collected from leaner non-sooting flames.
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Keywords: Soot; Nanoparticles; Raman spectroscopy

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doi:10.1016/j.proci.2010.07.077

1. Introduction

The morphology and chemical structure of soot particles has been thoroughly studied by a wide variety of techniques, like HR-TEM, X-ray diffraction, Raman and IR spectroscopy [1–6]. Soot particle structure is currently described as composed of carbon and hydrogen atoms in a ratio of H/C = 0.1–0.5. Carbon atoms form randomly oriented small crystallites consisting of 3–4 aromatic layers with a turbostratic stacking which causes the interplane distance to be generally larger than that of graphite crystals and which are linked together by van der Waals forces [7,8].

Soot precursor particles and incipient soot particles are much more elusive to analytical techniques. They can hardly be seen by HR-TEM microscopy and can decompose in the electron beam. They appear as translucent particles with extended regions of amorphous carbon where no crystallite are observed [1,7]. Precursor particles are also more deformable than aged soot as inferred by their low aspect ratio measured by Atomic Force Microscopy (AFM) when deposited by thermophoresis on mica substrates [9,10]. IR spectroscopy reveals a molecular-like character for soot precursors and incipient soot particles with spectra composed by sharp and narrow bands mainly due to aromatic and aliphatic bond vibrations [5,6,11]. In addition, medium range order is evidenced by Raman spectroscopy of soot precursor particles sampled in water, which shows the presence of the typical *G* and *D* bands measured in amorphous carbon compounds [12].

Spectral absorption measurements in the UV–visible range have been widely used to distinguish between soot precursor and soot particles [13,14]. Soot precursors were defined as particles absorbing in the UV spectral range whereas soot particles absorb light also in the visible and emit a laser induced incandescence signal. Following this classification, soot precursor particles were found in relevant amounts in flames with equivalence ratios (Φ) significantly lower than that corresponding to the onset of soot, in contrast to particle inception depicted by measuring particles with a dilution probe-Differential Mobility Analyzer (DMA) [15].

Figure 1 reports a sketch of the particle volume fraction determined in ethylene/air flames by light absorption compared to the soot volume fraction determined by laser induced incandescence [14,15]. In these flame conditions, the soot formation threshold is at about $\Phi = 2.0$. The onset of total particles, considered to be the condition in which the light absorption in the UV starts to noticeably increase above the gas phase background, is as low as $\Phi = 1.5$, below which, data uncertainty is too large to detect the presence of particles. DMA measurements show a threshold of particle formation intermediate between the previous two. Disagreement between the estimate of particle volume fraction from DMA and light absorption measurements has been ascribed to particle losses during sampling, or to the presence of absorbing species with mobility diameters smaller than the DMA detection limit of about 1.5 nm. Detailed chemical kinetic modeling results [14,16], also shown in Fig. 1, help explain the attribution of the measured particle volume fractions. The first onset of particles at $\Phi = 1.5$ is attributable to molecules of aromatic compounds with equivalent spherical size of 1–1.5 nm. The second onset of particles, measured by on-line DMA, coincides with the formation of clusters of aromatic molecules with an equivalent spherical size ranging from 1.5 to 3 nm. Primary soot particles and their aggregates are predicted to form at about $\Phi = 2.1$, in good agreement with the threshold of soot particles determined by light absorption in the visible and laser induced incandescence.

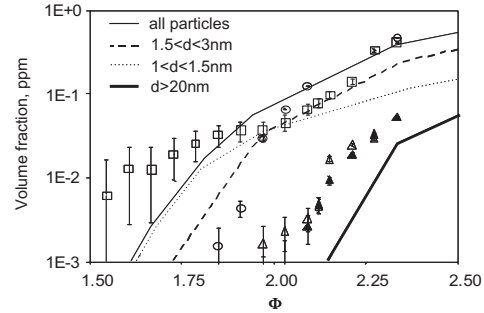

Fig. 1. Total particle volume fraction (f_v) determined by extinction in the UV and visible (squares), by on-line DMA measurements (circles), and volume fraction of soot particles determined by laser induced incandescence (empty triangles) and light extinction in the visible (filled triangle) versus equivalence ratio. Ethylene/air flames, cold gas flow velocity of 10 cm/s, height above the burner of 10 mm [15]. Lines: kinetic model predictions of f_v for species varying in size [16].

The above analysis evidences the need to shed light on the first steps of soot formation by improving knowledge of the composition of soot precursors and incipient soot particles. The objective of this paper is therefore to study the chemical composition of precursors and incipient soot particles produced in flames across the soot threshold.

Raman spectroscopy is a widely used technique to study carbonaceous compounds from graphite to amorphous carbon. Complementary to infrared spectroscopy, it is used to study vibrational transitions and medium-range order in the structure, and has been employed to characterize and compare soot produced in different conditions [2–4]. While infrared spectroscopy is more sensitive to the aliphatic vibrations, Raman scattering, excited by visible light, is highly sensitive to sp^2 bondings since π states can be resonantly

excited resulting in a much larger cross section than sp^3 sites.

Raman spectra of soot and amorphous carbon have been mostly studied in the wave number region of 1000–1800 cm^{-1}, where two main peaks are observed: one at about 1850 cm^{-1} denoted *G* peak for graphite and a second peak at 1350 cm^{-1}, called *D* peak for disorder. Another band at about 2700 cm^{-1}, denoted as 2*D*, is attributed to the second order of zone-boundary phonons, which give the first order *D* peak in defected graphite. The intensity of 2*D* band is much lower than *D* and *G* bands, and it has been only rarely used for the study of amorphous carbon [4].

Various methods have been developed to increase the intensity of Raman spectra, like Resonant Raman Spectroscopy, CARS and surface enhanced Raman spectroscopy (SERS). In particular, SERS is a powerful method able to provide intensity enhancements up to 14 orders of magnitude [17], which enables the detection of single molecules with the high structural selectivity provided by its Raman spectrum. SERS relies on the strongly increased Raman signal from molecules close to metallic nanostructures. The enhancement mechanisms, not yet fully understood, are usually divided into an electromagnetic field enhancement related to plasmonic excitation of the metallic nanostructures and chemical effects [18].

The strong enhancement of Raman signal operated by SERS has been exploited in this paper to study the 2*D* Raman band, in addition to first order Raman bands, on particles deposited in very low concentrations as isolated particles on SERS substrates either by thermophoresis or by electrophoresis down stream of a dilution probe-DMA sampling system, giving size-selected samples.

2. Experimental details

Particles were produced in premixed ethylene/air flames with cold gas velocity of 10 cm/s and C/O ratio varying from C/O = 0.61–0.72. Particle size distributions were measured by on-line DMA using a horizontal-tube dilution probe to suppress particle coagulation, cool the sample to room temperature, and transport it to the inlet of a TapCon 3/150 DMA equipped with a Faraday Cup Electrometer detector. N_2 dilution flow through the probe was set to a dilution ratio of 1200, which suppressed particle growth processes in the sampling system since the measured size distribution showed no change in shape with a further increase of the dilution. More details are reported in a previous work [15].

Particles were deposited on SERS substrates either by thermophoretic or electrophoretic sampling.

Thermophoretic sampling was performed to collect particles directly from the centerline of the flame at a specific height. We rapidly inserted (sample time = 30 ms) the substrates oriented face-down horizontally, parallel to the burner surface (rather than parallel to the flame streamlines) to enable better spatial resolution. Also, the horizontal orientation used for thermophoretic sampling is more commensurate with the flow conditions and flame perturbations that arise near the surface of the horizontal-tube dilution probe used to collect the size-selected samples.

Electrophoretic sampling of size-selected particles was performed on-line downstream of the electrostatic classifier of a DMA where a fixed voltage was applied to the electrodes for the duration of the sampling. Monodisperse particles exiting the electrostatic classifier were deposited on SERS substrates by means of an electrostatic precipitator (EP) similar to the one described in detail by Dixkens and Fissan [19]. The monodispersity of the selected aerosol is limited by the classifier geometry and diffusion effects within the DMA which worsen with decreasing diameter. For the selected size intervals $\Delta d/d = 0.2$ [20]. The electrostatic sample time ranged from 8 to 30 min, depending on the particle concentration, specific for the given flame condition, in order to achieve a deposition density of about 100 particles/μm^2, assuming 100% collection efficiency. The voltage applied to the EP was 8 kV.

SERS spectra were measured with a Raman microscope, consisting of an inverted microscope, equipped with a Raman probe. The Raman probe is composed of a frequency-doubled ($\lambda_0 = 532$ nm) Nd-YVO laser tightly focused onto the sample by an Olympus oil-immersion infinity corrected objective lens (100X, 1.4 N.A.). The laser power density was reduced to 30 W/mm^2 by means of neutral density filters in order to avoid particle damage. Back-scattered light from the sample was collected and collimated by the same focusing objective, and, following back the same path as the incident Raman probe, reached finally the notch filter. After passing through a 50 μm pinhole, used for confocal photons collection, the radiation was focused onto the entrance slit (50 μm) of the spectrometer (1800 lines/mm holographic grating) coupled to a CCD camera (1024 × 100 pixels). The signals measured in the range of 500–2200 cm^{-1} were integrated for 10 s and those measured in the range of 2300–3400 cm^{-1} for 20 s in order to achieve a signal to noise ratio larger than 100.

Ag nanoparticles, used as SERS-active substrate, were prepared following the procedure described in [21]. Briefly, 200 mL of a 10^{-3} M $AgNO_3$ solution was brought to boiling conditions, and 4 mL of a 1% sodium citrate solution were added one drop at a time under vigorous stirring. The resulting solution was kept boiling

for 1 h, during which it becomes dark grey. The resultant colloidal mixture was cooled at ambient conditions to room temperature. This procedure produces spherical Ag nanoparticles, with an average diameter around 40 nm [22], and assures no contamination of the substrate with carbon-containing material. The colloidal solution was finally dropped on the center of glass microscope coverslips, which were dried at atmospheric pressure in a clean chamber.

3. Results and discussion

Figure 2 reports three representative size distributions measured by on-line DMA in different flame conditions. Generally, three particle size modes are observed by on-line DMA in premixed flames. The smallest mode, which we will call $mode_I$, has a modal diameter of about 2 nm that varies in size only slightly with C/O. $Mode_I$ is the only mode observed in flame conditions near the onset of particle formation (C/O < 0.65 all heights), and it is observed also in richer flames at low height. In flames richer than about C/O ~ 0.65 and C/O ~ 0.7 larger size modes, $mode_{II}$ and $mode_{III}$, appear. Both of which vary with flame conditions growing to larger sizes and broader distributions with increasing height and C/O. $Mode_{III}$ extends to sizes much larger than 10 nm, and is associated with soot particles since it appears only in flame conditions which present an incandescence signal and absorption/fluorescence in the visible [15].

We collected various particle samples for SERS measurements in order to comparatively analyze the chemical composition of $mode_I$, $mode_{II}$, and $mode_{III}$ particles. Two sampling procedures were used to collect particles on the SERS substrate: thermophoretic sampling on cold substrates inserted into the flame and electrophoretic sampling at room temperature, depositing particles of a specific size-selected by on-line DMA.

Figure 3 reports typical Raman spectra measured for particles having sizes selected by the

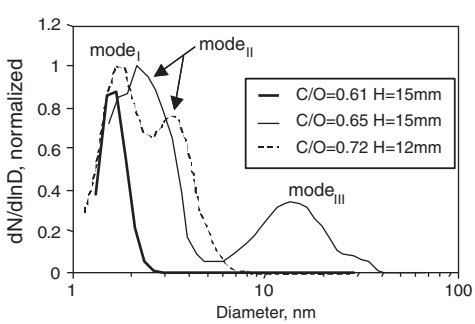

Fig. 2. Normalized size distribution function measured in premixed ethylene/air flames.

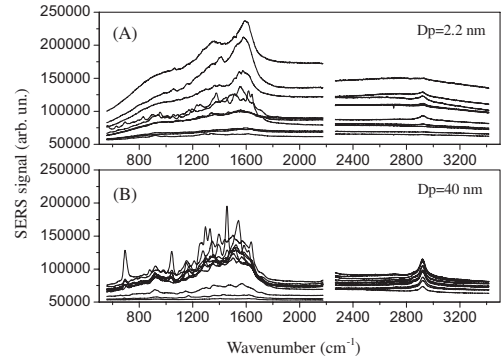

Fig. 3. (A) SERS spectra of particles with size of $d = 2.2$ nm sampled from C/O = 0.61 flame. (B) $d = 40$ nm particles sampled from C/O = 0.72 flame. Different curves are repeated measurements in different points of the SERS substrate.

DMA. The various curves are repeated measurements in the same sample obtained measuring various points on the substrate. The spectra present the typical *D* and *G* bands of carbon materials, respectively at about 1350, 1580 and their overtone and combination bands above 2400 cm^{-1}. In most cases, we observed a rather large intensity variability in repeated measurements, which is often reported in the literature for SERS measurements [23]. It can be due to various effects. "Blinking" is the phenomenon of intensity variation of the signal over time; it is an intrinsic drawback of SERS due to the nature of the enhancement effect, which depends strongly on the coupling between the metallic colloid and particles. It occurs in regions on the order of a few nanometers called "hot spots", which are not stable or predictable. The very high spatial resolution of enhancement also makes the SERS signal sensitive to interparticle and intraparticle dishomogeneity, so that while a conventional Raman scattering would give Gaussian broadened bands, in the case of SERS, when a given atomic group gets close to the silver particle surface the Raman signal related to its vibrations is selectively amplified, giving a peak in the spectrum. The presence of sharp peaks can be therefore interpreted as evidence for the composite character of the *G* and *D* band, which are inhomogeneously broadened [23]. A further factor that must be taken into account is the SERS selection rule which establishes a strong enhancement for those bondings whose vibrations are orthogonal to the metallic surface, so that line oscillations can also reflect different orientations of similar particles on the metallic substrate [24]. It is relevant to note that in Fig. 3 sharp peaks are more abundant in the spectra of the larger particles. Since the size of the 2.2 nm particles is on the same order as that of the "hot spot", their measured spectra is an average response from the

entire particle. Instead, for particles much larger than the "hot spots", each spectrum provides information on local structural units, or the main bonds on the surface of the particle in the vicinity of the "hot spot".

Most of the sharp peaks can be attributed to PAH-like structural units [25,26]. Peaks at 1150 and 1450 cm^{-1} have been attributed to transpolyacetylene like structural units [27], but the peak at 1450 cm^{-1} has been also associated to five-sided or seven-sided graphene rings [28], or, instead, to linearly aggregated six-membered rings [26].

Kudelski and Pettinger [29] reported that averaged spectra, although presenting reminiscence of fluctuations, are identical to the spectra measured during periods without large fluctuations or measured with a standard Raman spectrograph. On this bases, to examine the relative behavior of the G and D bands in different particle samples, we averaged 10 spectra for each sample, enough to significantly reduce fluctuations. The averaged curves converged to within 10% by averaging more than six spectra for the second order spectra, and in the case of the first order spectra, averaging more than eight spectra gave results converging to within 20%.

Figure 4 compares the averaged spectra, after a linear background subtraction, of the sample collected in the C/O = 0.61 flame by thermophoretic sampling (Fig. 4A) with that collected selecting only $d = 2.2$ nm particles (Fig. 4B). Each spectrum is the convolution of different bands determined by fitting the curve with Lorentzian and Gaussian curves. In the region between 1000 and 2000 cm^{-1}, six bands can be observed. Following Sadezky et al. [4], the band identification is: the G band is centered at about 1580 cm^{-1} and corresponds to the mode of ideal graphitic lattice; sp^2 carbon in linear chain also contribute to this mode, which is then representative of all sp^2 carbon in the particle. The $D1$ band is centered at about 1350 cm^{-1}; it is Raman active only in disordered graphitic lattices and can be attributed to graphene layer edges. The $D2$ band, centered at about 1620 cm^{-1}, is attributed to surface graphene layer in disordered graphite. The $D3$ band, centered at about 1500 cm^{-1} is related to amorphous carbon. Finally, the $D4$ band, centered at about 1200 cm^{-1} is attributed to disordered graphitic lattices. Another band centered at 1150 cm^{-1} is also observed, whose attribution is not certain. In agreement with Sadesky et al. [4], the best fit, based on the Levenberg–Marquardt method, was performed using a Gaussian lineshape for the $D3$ band and Lorentian curves for all other bands. The quality of fit is reported in the figures in terms of the reduced chi-squared (χ^2/DoF) and R-square correlation coefficients.

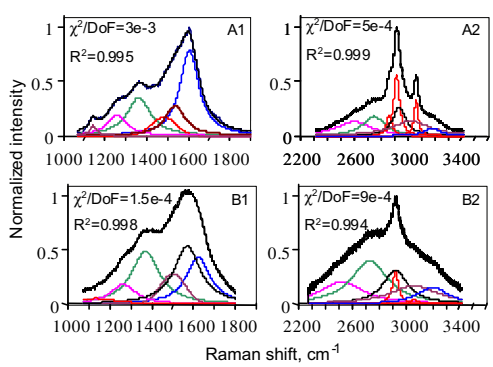

Fig. 4. SERS spectra of the sample collected in the flame with C/O = 0.61 by thermophoretic sampling (A) and at the exit of the DMA selecting the particle diameter $d = 2.2$ nm (B). Continuous lines report the results of the line fit.

The spectra of amorphous carbon in the region of 2000–2900 cm^{-1} are considered as the superposition of five bands due to second order or combinations of the bands appearing in the previous spectral region and were always fitted by Lorentian lineshape [4]. The band appearing at about 2500–2600 cm^{-1} is attributed to the second order of the $D4$ band, $2 * D4$. The $2 * D1$ band is centered at 2720–2750 cm^{-1} and is the second order of the $D1$ band; this band is generally the most intense observed in amorphous carbon. The band at about 2920 cm^{-1} is attributed to the combination of the G and $D1$ band, $G + D1$. The second order of the $D3$ band is observed at 3000–3060 cm^{-1}, $2 * D3$. The band centered at 3150–3210 cm^{-1} is finally attributed to the second order of G or $D2$ band, $2 * G$, $2 * D2$. Figure 4 clearly shows that three additional sharp peaks at 2850, 2920 and 3050 cm^{-1} are detected in our SERS spectra. Therefore, we fitted the spectra with the superposition of eight Lorentian curves.

The spectra of the samples collected in the flame with C/O = 0.61, where the size distribution function measured by the DMA is composed only by $mode_I$ particles, well reproduce all the features reported for amorphous carbon. Nevertheless, some differences appear between the spectrum of Fig. 4A and B. The particles collected by thermophoresis present an intense band in correspondence of the $D2$ band at about 1600 cm^{-1}. However, the most interesting Raman features, that highlight differences between samples can be observed in the second order region of the spectrum: the bands centered at 2850, 2920 and 3050 cm^{-1} which are very strong and have a narrower width, less than about 50 cm^{-1}, than the other bands of amorphous carbon. Bonding vibrations at 2920 and 2850 cm^{-1} can be assigned to Raman active sp^3 CH_2 vibrations [5,6,11,25,30,31]; they were previously measured also in infrared spectra of particles collected in similar flames by water-based sampling [11]. The peak at 3050 cm^{-1} is instead indicative of aromatic

C–H stretching vibration [5,6,25]. We exclude the attribution of this signal to small PAH molecules since they cannot be detected by SERS (without coatings on the SERS substrate) [32]. The difference between SERS spectra of thermophoretic sampled material and of size-selected particles are probably due to macromolecules of aromatic compounds, which are present in flame together with $mode_I$ particles and were collected by thermophoretic sampling, but did not pass through the DMA classifier. Such species have a molecular-like character with vibrational transitions similar to those observed for PAHs: CH stretching at 3050 cm^{-1}, aromatic C–C stretching at about 1600 cm^{-1} and sp^3 CH_2 vibrations at 1850 and 1920 cm^{-1}. Those species would correspond to the molecules with equivalent size range of $1 < d < 1.5$ nm predicted by the model discussed in the introduction and reported in Fig. 1.

Figure 5 reports the SERS spectra of $mode_I$ and $mode_{II}$ particles sampled from the C/O = 0.65 flame ($\Phi = 1.97$), respectively with $d = 2.4$ and 4 nm. The SERS spectrum of smaller particles, $d = 2.4$ nm, is very similar to the one discussed earlier for particles with comparable size produced in the leaner flame (Fig. 4B). The SERS spectrum of $mode_{II}$ particles is rather similar as well, but two main differences between the $mode_I$ (Fig. 5A) and $mode_{II}$ (Fig. 5B) particles can be observed. The band in correspondence with the G peak in $mode_{II}$ particles becomes broader and widens towards smaller frequencies and $D3$ band at 1530 cm^{-1} increases. The second order spectrum is more symmetric than in the previous cases with a higher contribution of the broad $2 * D3$ band at about 3050 cm^{-1}. The shift of the G peak caused by additional contributions at smaller frequencies, below that of graphite at 1580 cm^{-1}, is indicative of vibrations from sp^2 C–C with increased bond length. It likely results from increases in the number of distorted aromatic rings or rings of others orders with respect to a regular aromatic structure or by olephinic chains [30]. Probably, it is the presence of distorted aromatic rings, which would justify also the increase of the bands ascribed to disorder, $D3$ and $2 * D3$.

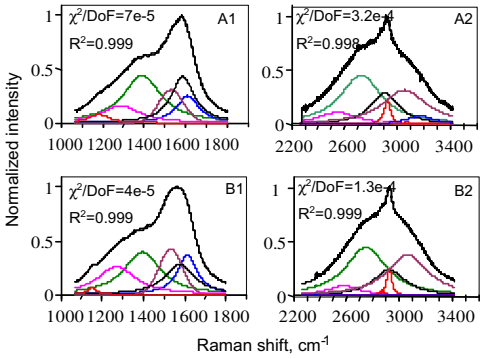

Fig. 5. SERS spectra of the sample collected in the flame with C/O = 0.65 at the exit of the DMA selecting the particle diameter $d = 2.4$ nm (A) and $d = 4$ nm (B). Continuous lines report the results of the line fit.

In the richer flame condition, C/O = 0.72 ($\Phi = 2.18$), many sharp lines are superimposed on the first order G and D bands. One explanation for their appearance is that the multiplicity of sharp lines, as reported in Fig. 3, could be evidence of bonds present on the surface of the particle vibrating preferentially orthogonal to the substrate surface. According to this reasoning, these lines are more evident in spectra of the particles collected from the richest flame studied (Fig. 3b) because of their higher aspect ratio [9,10]. Also, the absence of sharp lines in SERS spectra of particles formed in leaner conditions can be related to the particle deformability and shape on the SERS-active substrate. In fact, previous work demonstrated that particles in non-sooting flames assume a flat shape when collected on substrates [9,10], consequently, most of their vibrations could be parallel to the metallic surface and not satisfy the SERS selection rules.

Averaging multiple spectra reduces the intensity of the sharp lines, but the residual presence of structures limits the accuracy of the fitting procedure.

The analysis of first order spectrum do not evidence a high degree of order even in the case of large soot particles, $d = 40$ nm, since all of the D lines are significant. Actually, for structures with small crystallites size, less than 2 nm, which is the case of nascent soot particles [7], the D band intensity indicates the probability to find aromatic rings, consequently its increase evidence increasing aromaticity [30]. In this regime, the ratio of the D and G bands intensities $I(D)/I(G)$ is, in fact, proportional to the square of the aromatic island size or in plane correlation length, La. According to the equation proposed by Ferrari and Robertson [30] La is on the order of 1–1.5 nm for all particles analyzed.

In the second order spectra, the line dishomogeneities are completely absent, and the spectra are very reproducible and well-fit by the superposition of eight Lorentian curves. The spectra measured for the $d = 2.7$ nm particles present two different shapes, which were averaged and graphed separately in Fig. 6 (A2-1 and A2-2); their main difference regards the intensity of the lines. The spectrum reported in Fig. 6A2-1 is quite similar to the spectra of particles produced in non-sooting flames, while the spectrum in Fig. 6A2-2 is much more similar to the one for $d = 40$ nm particles, Fig. 6B2. All of the particles in the C/O = 0.72 flame present a strong band positioned at 2920 cm^{-1} that is superimposed on a relatively smaller background, which resemble the spectra measured in the previous non-sooting flames. The strong peak is instead similar to the one

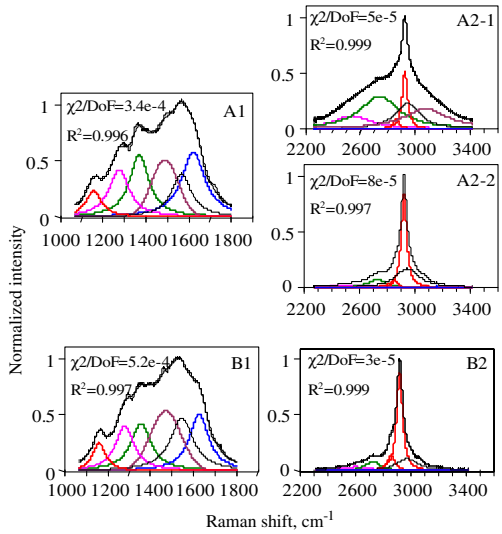

Fig. 6. SERS spectra of the sample collected in the flame with C/O = 0.72 at the exit of the DMA selecting the particle diameter $d = 2.7$ nm (A) and $d = 40$ nm (B). Continuous lines report the results of the line fit.

measured in the sample collected by thermophoresis and reported in Fig. 4A. As previously discussed, the extremely constant position and narrow lineshape of the 2920 cm^{-1} line strengthens the attribution of this peak to vibrations involving H atoms rather than the combination of the G and $D1$ band since both the G and $D1$ peaks are much wider and have variable peak positions. Its strong intensity measured in SERS spectra of particles produced above the soot threshold (C/O = 0.72 flame) is not completely clear; it is probably not dependent on the concentration of H atoms in the particles since the H/C ratio is usually reported to decrease in larger soot particles [7]. The intensity of this peak may depend on the orientation of the vibrations of H atoms on the particle surface close to the SERS "hot spots" and would therefore be enhanced for more spherical particles, as discussed for the sharp lines in the first order spectra. Another possible explanation that cannot be excluded is that the particles undergo more interaction with hydrogenated species possibly because of an higher surface reactivity.

The second harmonic region of the D and G bands is particularly interesting since it is sensitive to structural changes along the c-axis in crystallite and can be used to obtain information on three-dimensional ordering. Lespade et al. reported a correlation between the wideness of the $2 * D1$ band at 2700 cm^{-1} and the d_{002} interplane spacing [33]. This parameter has been found to decrease as the interplane spacing decreases in various amorphous carbon materials, reaching a minimum on the order of 50 cm^{-1} in correspondence with $d_{002} = 3.40$ Å, and to rise again in highly orga-

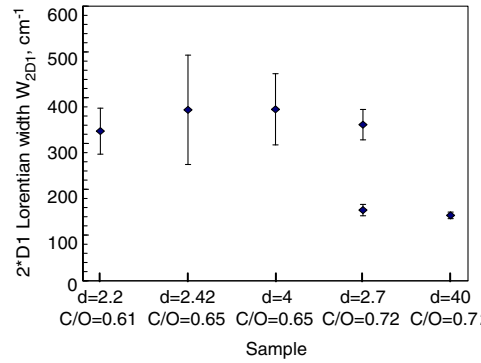

Fig. 7. Width of the Lorentian $2 * D1$ curve centered at about 2730 cm^{-1}. Two separate points are plotted for the $d = 2.7$ nm C/O = 0.72 sample as in Fig. 6 A1 and A2.

nized carbon until the band at 2700 cm^{-1} finally splits up in graphitic lattices.

The width of the $2 * D1$ Lorentian curve, W_{2D1}, measured for the various particles investigated is reported in Fig. 7. $Mode_I$ particles, with $d \sim 2$ nm, have a very large width of more than 250 cm^{-1}. Even considering the uncertainty of the data, this value is much higher than the values reported by Lespade et al. [33] who found a maximum of about 100 cm^{-1} in correspondence of d_{002} of about 3.55 Å. As a consequence, $mode_I$ particles present a very poor, if any, three-dimensional ordering. W_{2D1} remains significantly high also for $mode_{II}$ particles, $d \sim 4$ nm, suggesting that also these particles do not have strong interplane interactions. The three-dimensional ordering becomes appreciably higher for particles collected from soot-forming flames, since a lower W_{2D1} is measured (Fig. 7) even though particles as large as $d = 40$ nm still have $W_{2D1} > 100$ cm^{-1}. This observation suggests that crystallites are present in the particles, but the interlayer d_{002} is still larger than 3.50 Å due to the presence of defects in the graphitic planes.

It is interesting to note that in the case of particles with $d = 2.7$ nm sampled in the flame C/O = 0.72, the W_{2D1} parameter confirms two different particle types: one similar to particles collected in non-sooting flames, with a broader Lorentian band and a second particle type with a higher three-dimensional ordering, similar to the 40 nm $mode_{III}$ particles. The particles with higher three-dimensional ordering are probably composed of stacking of PAH-like species even though they present lateral sp^3 CH_2 bonding.

4. Conclusions

In this work we used SERS for the first time to study the composition of particles produced in ethylene/air premixed flames across the soot

threshold limit, in flames conditions producing three modal values of the size distribution function roughly at 2, 4 nm and $d > 10$ nm.

The strong enhancement produced by the SERS substrates enabled the measurement of Raman scattering from monodisperse isolated particles in very low concentrations and of the second harmonic of the *D* and *G* bands, which give complementary information to the more used first order spectra. The strong spatial resolution of SERS enhancement occurs only in "hot spots" on the order of a few nanometers, which causes strong oscillation in the position and intensity of sharp peaks in the spectra, revealing a complex, inhomogeneous particle composition. While this effect can be exploited for a detailed determination of particle surface composition, in this paper, we analyzed averaged spectra to study *D* and *G* bands typical of carbon materials.

Flames with $C/O < 0.65$ produce only $mode_I$ particles, which present the typical bands of amorphous carbon with no evidence of three-dimensional ordering of graphitic planes. Other macromolecules with vibrational transitions similar to those observed for PAHs with side chains are present in the flame and observed in the thermophoretic samples. In the bimodal non-sooting flame, $mode_I$ particles are similar to those found in the leaner conditions, while $mode_{II}$ particles present larger aromatic islands but also higher degree of disorder, probably due to the presence of distorted or five order rings. The formation of crystallites by the stacking of aromatic island was not evidenced for either $mode_I$ or $mode_{II}$ particles in non-sooting flames.

An increase of three-dimensional ordering was observed in both $mode_{II}$ and $mode_{III}$ particles only in soot-forming flames. Nevertheless, particles as large as $d = 40$ nm are far from being graphitized since their crystallite interplane distance d_{002} is still larger than 3.50 Å and a relevant signal from surface CH_2 aliphatic chain is detected.

Size-selected SERS measurements demonstrate that particles with the same size produced in sooting and non-sooting flame conditions have distinctly different composition.

SERS is a very promising technique for the study of particles composition and evolution in flames, allowing a detailed analysis of internal ordering and composition of isolated particles in samples with extremely low concentrations. Also, thanks to the large enhancement of surface vibrations and molecular specificity, SERS has great potentiality to study soot surface reactions.

Acknowledgement

We thank Rocco Pagliara for constructing the electrostatic precipitator.

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